

UV Completion: Asymptotic Safety Fixed-Point Search in MQGT-SCF

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Abstract

We perform an asymptotic safety fixed-point search in the Bridge D truncation (2 real scalars with 1/24 diagonal quartics, 1/4 portal) extended with Narain-Percacci-Wirth gravitational corrections. The gravitational coupling is fixed at the corpus’s reconstructed 33-coordinate joint fixed point: $g_N^* = 1.7621271$. **Result: No non-Gaussian fixed point found.** All 91 trajectories from the positivity cone flow to the Gaussian fixed point ($\lambda = g = 0$) in the UV. Gravity drives the matter sector to Gaussian in the UV. Consistent with corpus statement: “matter sector is Gaussian at joint FP.”

1 Introduction

The MQGT-SCF framework [?] extends the Standard Model with two real scalar singlets Φ_c and E , coupled to a locally sourced GKSL measurement container. The Bridge D verification [?] established the scalar sector’s positivity cone and UV behavior. Here we extend this with asymptotic safety gravitational corrections [?] to search for a non-Gaussian UV fixed point.

2 Method

2.1 Truncation

Scalar sector: 2 real scalars, diagonal quartics 1/24, portal 1/4. Matter betas (Machacek-Vaughn):

$$\beta_{\lambda_1} = \frac{3(\lambda_1^2 + g^2)}{16\pi^2}, \quad (1)$$

$$\beta_{\lambda_2} = \frac{3(\lambda_2^2 + g^2)}{16\pi^2}, \quad (2)$$

$$\beta_g = \frac{g(\lambda_1 + \lambda_2) + 4g^2}{16\pi^2}. \quad (3)$$

Gravitational corrections (NPW, arXiv:0906.1974):

$$\Delta\beta_{\lambda_i} = -\frac{g_N}{\pi} \lambda_i, \quad (4)$$

$$\Delta\beta_g = -\frac{g_N}{\pi} g. \quad (5)$$

Fixed $g_N^* = 1.7621271$ (from 33-coordinate flow [?]).

2.2 Positivity Cone

$\lambda_1 > 0, \lambda_2 > 0, g^2 < \lambda_1 \lambda_2 / 9$ (copositivity).

2.3 Search

91 initial conditions in positivity cone ($\lambda_{1,2} \in \{0.01, 0.03, 0.1, 0.3, 1.0\}, g \in \{0.001, 0.003, 0.01, 0.03, 0.1\}$). RG integration from IR to UV: $t_{\max} = 10, n_{\text{steps}} = 2000$. Grid search for approximate FPs ($|\beta| < 0.05$).

3 Results

Metric	Value
g_N^*	1.7621271
Initial conditions tested	91
Trajectories surviving to UV	91/91
UV endpoint	Gaussian ($\lambda, g \rightarrow 0$)
Non-Gaussian FPs found	0

All 91 trajectories survive to UV and flow to the Gaussian fixed point. The gravitational correction term $-(g_N/\pi)\lambda$ drives all matter couplings to zero in the UV.

4 Interpretation

1. **Gaussian UV completion:** The scalar sector is asymptotically free (Gaussian) in the UV. No non-Gaussian fixed point exists in this truncation.
2. **Gravity drives matter to Gaussian:** The negative gravitational correction to scalar quartics $(-g_N/\pi\lambda)$ dominates the positive matter self-interaction at high scales, driving all couplings to zero.
3. **Consistency with corpus:** “matter sector is Gaussian at joint FP” — confirmed.
4. **No UV completion barrier:** The scalar sector does not obstruct the asymptotic safety program at this truncation level.

5 4-Scalar Extension

A 4-scalar truncation ($g_N^* = 1.7621271$) was also tested. Result: **No non-Gaussian FP found; all UV flows $\rightarrow$ Gaussian.** Consistent with 2-scalar result.

6 Code & Reproducibility

Script: `bridge_d_uv.py` (Phase 3 UV Completion). 4-scalar script: `bridge_d_uv_4scalar.py`. Output: `bridge_d_uv_4scalar_output.json`. Dependencies: `mpmath, numpy, csv`. Computational cost: ~ 1 second.

7 Conclusion

The scalar sector of MQGT-SCF is asymptotically free in the UV. The gravitational correction term $-(g_N/\pi)\lambda$ dominates matter self-interactions at high scales, driving all couplings to zero. This confirms the corpus statement that the “matter sector is Gaussian at joint FP” and removes the scalar sector as a UV completion barrier.